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Functions & Recursions

A function is a group of statement performing a specific task when a program gets bigger in size and it's complexity grows. It's gets difficult for a program to keep track on which piece of code doing what!

A function can be reused by the programmer in a given program any number of.

Example and Syntax of a function

The Syntax of a function looks as follows:

```
def func1():
```

```
    print("Hello")
```

The function can be called any number of times, any where in the program

• Function Call

Whenever we want to call a function, we put the name of function followed by the parenthesis as follows:

```
func1()
```

This is the function call.

• Ques: Write a program to greet a user with "Good day" using function.

• Type of Function in Python

➤ There are two types of functions in Python:

- Built-in Python Function (Already present in Python)
- User defined functions (Defined by the user)

Examples of built in function includes `len()`, `print()`, `range()` etc.

The `func1()` function we defined is an example of user-defined function.

• Function with Arguments

A function can accept some value if can work with we can put these values in the parentheses.

A function can also return value as shown below:

```
def greet(name):  
    gr = "hello" + name  
    return gr
```

```
a = greet("Harry")
```

a will now contain Hello Harry

• Default Parameter Value

We can have a value as default as default argument in a function.

if we specify `name = "stranger"` in the line containing `def`

This value is used when no argument is passed.

Examples:

```
def greet(name = "Stranger"):  
    greet() # name will be "Stranger" in function  
            body (default)  
  
    greet("Mahip") # name will be "Mahip" in  
                   function body (passed).
```

★ Recursion

Recursion is a function which calls itself.
It is used to directly use a formula mathematical as a function.

Example:

$$\text{factorial}(n) = n * \text{factorial}(n-1)$$

This function can be defined as follows:

```
def factorial(n):
```

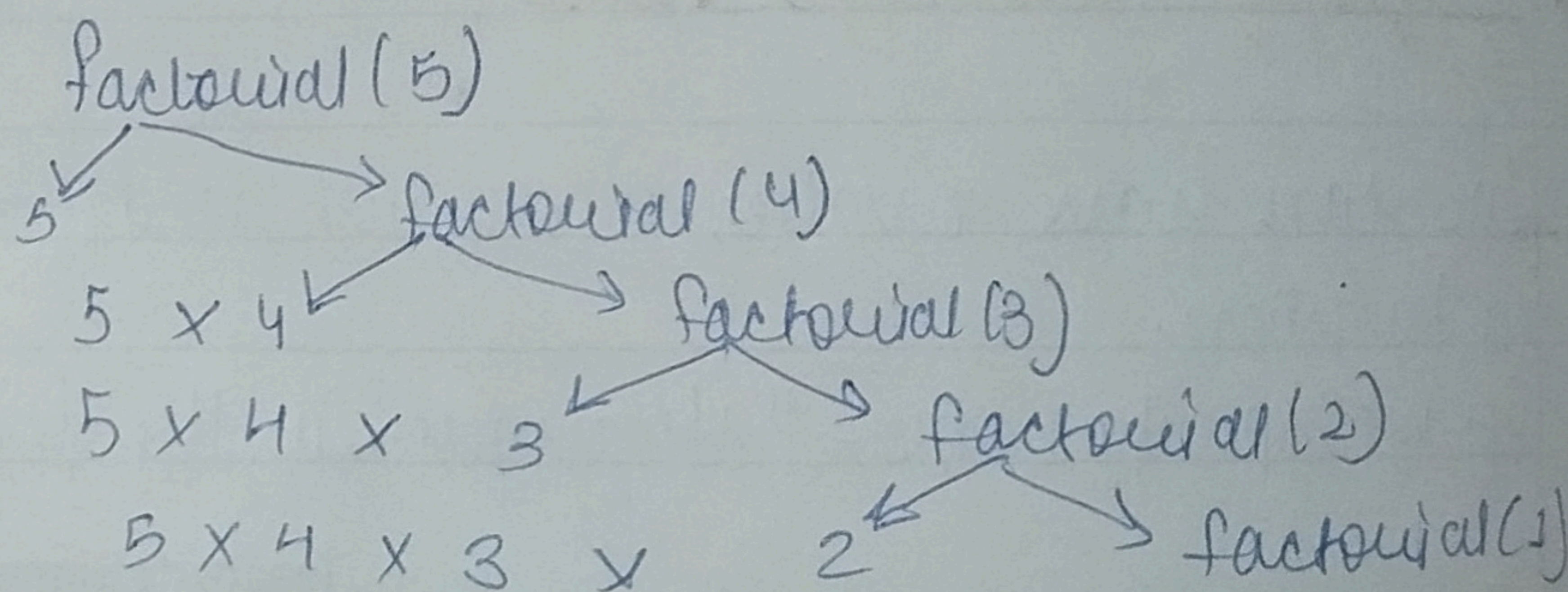
```
    if (n==1 or n==0):  
        return 1
```

```
    else:
```

```
        return n * factorial(n-1)
```

```
n = int(input("Enter a number value: "))
```

```
print("The factorial of a", n, "will be =", factorial(n))  
factorial(n)
```



5 x 4 x 3 x 2 x 1

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Practice Set

- 1) Write a Program using functions to find greatest of three numbers.
- 2) Write a python Program using function to convert Celsius to Fahrenheit.
- 3) How do you prevent a python print() function to print a new line at the end.
- 4) Write a recursive function to calculate the sum of first n natural numbers.
- 5) Write a Python Function to print first n lines of the following patterns:

```
***  
**  form = 3  
*
```
- 6) Write a python Function which converts inches to cms.
- 7) Write a python Function to remove a given word from a list and strip it at the same time.
- 8) Write a python Function to print multiplication table of a given number.